

Overview of the watsonx experience

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The watsonx experience is a secure and collaborative environment where you can access your organization's trusted data, automate AI processes, and deliver AI in your applications. The IBM watsonx.ai services provide a studio of integrated tools for building generative AI and machine learning solutions. The IBM watsonx.governance service provides end-to-end monitoring for AI solutions to accelerate responsible, transparent, and explainable AI workflows.

Use cases [↗](#)

AI engineers, data scientists, and AI risk and compliance officers can accomplish the following goals with watsonx.ai and watsonx.governance:

Build generative AI solutions

Build solutions that include prompts, agents, retrieval-augmented generation (RAG) patterns, and other capabilities of foundation models. Choose from IBM models, third-party models, open source models, or import custom foundation models. Tune foundation models to customize your prompt output.

Build machine learning solutions

Build models with open source frameworks and code-based, automated, or visual data science tools. Manage and automate the model lifecycle with integrated tools and runtimes to train, validate, and deploy machine learning models.

Govern AI

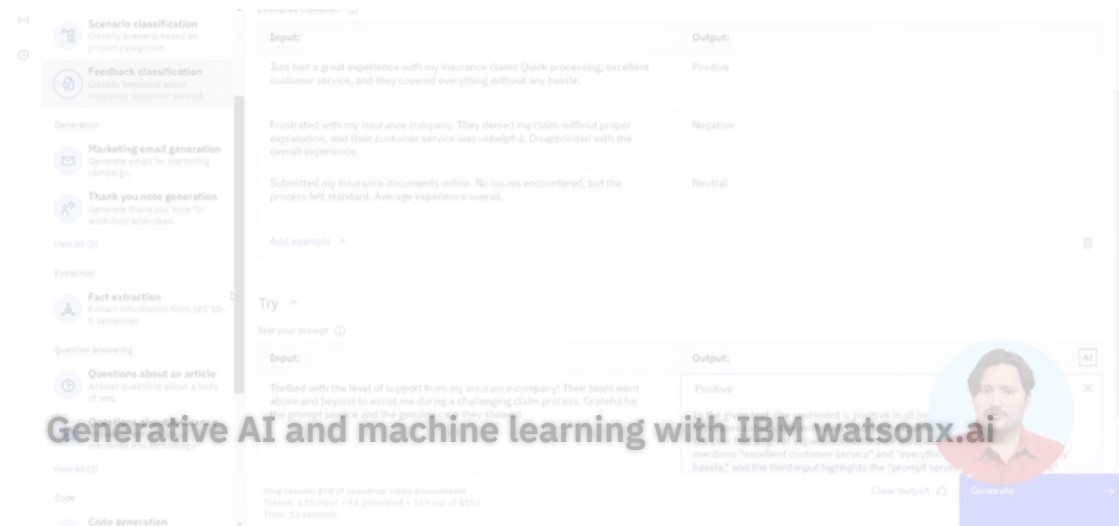
Track the detailed history of AI models, assess risks, and evaluate model output to help ensure compliance.

Watsonx.ai [↗](#)

Watsonx.ai includes APIs and tools for building AI solutions, deployed foundation models for generative AI, and the hardware and software resources for computing and inferencing.

Watch this short video that introduces watsonx.ai on IBM Cloud. The video begins on the watsonx home screen. The user selects **Open Prompt Lab** to start working in the Prompt Lab.

This video provides a visual method to learn the concepts and tasks in this documentation.



Ways to prepare data and build AI solutions [🔗](#)

For most tasks, you can choose between writing code and working with tools in the UI. Your watsonx.ai tools are in collaborative workspaces called projects.

You can prepare data and build AI solutions in the following ways:

Prepare data for AI

- Refine and visualize your data files or data tables in remote data sources with Data Refinery.
- Generate synthetic structured data for training machine learning models with Synthetic Data Generator.
- Generate synthetic unstructured data for tuning foundation models or testing gen AI solutions with the synthetic data generation API.
- Vectorize your unstructured data for RAG patterns with vector indexes.

See [Preparing data](#).

Build generative AI solutions

- Write generative AI solution code with Python SDKs, REST APIs, or Node.js.
- Experiment with generative AI prompts in the Prompt Lab.
- Automate RAG patterns with AutoAI.
- Tune foundation models for your use case with Tuning Studio.
- Build AI agents with Agent Lab.

See [Developing generative AI solutions](#).

Build machine learning models

- Automatically generate predictive model candidates with AutoAI.
- Create machine learning model training flows with SPSS Modeler.
- Write Jupyter notebooks to train models in Python or R.
- Solve optimization problems with Decision Optimization.
- Automate the machine learning lifecycle with Orchestration Pipelines.

See [Data science solutions](#).

Deployed foundation models

IBM watsonx.ai has a range of deployed large language models for generative AI. The deployed foundation models include open source models and IBM foundation models. You can also deploy your own custom foundation models or deploy an on-demand model. You can customize foundation model behavior by tuning a foundation model.

For a list of supported foundation models that are deployed in watsonx.ai, see [Supported foundation models](#).

Usage resources

Depending on your service plans, you might have a set amount of usage resources per month, per year, or you might be billed for the resources that you consume.

When you run tools or host models on watsonx.ai, you consume the following types of resources:

Compute usage

When you run jobs, notebooks, experiments that train or tune models, or deployments, your compute resource usage is calculated based on the rate for the runtime environment and its active duration. Compute resources include the appropriate hardware and software that are specific for the workload.

Inferencing usage

When you run inferencing against foundation models, your inferencing usage is calculated as the sum of the tokens in the prompt input and output text multiplied by the rate for the foundation model. Tokens are basic units of text.

Model hosting

When you deploy a custom foundation model or a deploy-on-demand foundation model, you are charged an hourly rate. Billing rates are according to model hardware configuration and cover both hosting and inferencing the model. Resource measurements begin when the model is successfully deployed and continues until the model is deleted.

Text extraction

When you use text extraction to convert document files into an AI model-friendly file format, resources for processing each page are measured.

Learn more about usage and billing:

- [Billing details for watsonx.ai Studio tools](#)
- [Billing details for generative AI assets](#)
- [Billing details for machine learning assets](#)

Watsonx.governance

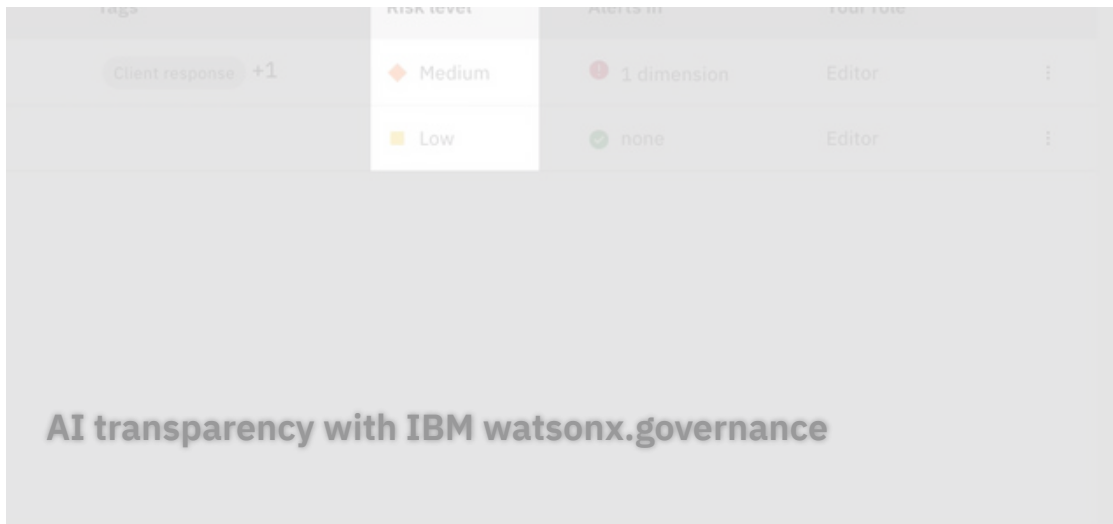
Watsonx.governance includes tools for governing models and the usage resources for evaluating and explaining models.

[Deployment differences](#) AI Governance capabilities differ depending on your deployment environment:

- Watsonx.governance on IBM Cloud provides most AI governance capabilities. You can integrate the IBM OpenPages service to enable the Governance console. All solutions are available (licensing is required).
- Watsonx.governance on Amazon Web Services (AWS) provides the Governance console with the Model Risk Governance solution.

Watch this short video that introduces watsonx.governance.

This video provides a visual method to learn the concepts and tasks in this documentation.



Ways to govern AI [🔗](#)

You can use watsonx.governance APIs and tools to govern AI in the following ways:

Monitor and evaluate AI assets

- Monitor machine learning model output and explain model predictions.
- Evaluate and compare generative AI prompts.

Your watsonx.governance model monitoring and evaluation tools are in projects and deployment spaces.

See [Evaluating AI assets](#).

Track and document AI use cases

View model lifecycle status, general model and deployment details, training information and metrics, and deployment metrics with AI use cases.

See [Governing assets in AI use cases](#).

Manage governance activity

- Sync data from factsheets with the Governance console.
- Extend governance capabilities with workflows and other compliance tools with the Governance console.

You must integrate with the watsonx Governance console from IBM OpenPages.

See [Managing risk and compliance with Governance console](#).

Share governed AI agents and tools

Register, manage, evaluate, and reuse AI-powered agents and tools in the governed agentic catalog.

See [Governed agentic catalog](#).

Manage and share AI guardrail policies

Create, manage, and reuse guardrail policies in an inventory.

See [Configuring guardrails](#).

Usage resources [↗](#)

When you run model evaluations and explanations with watsonx.governance, you consume resources. Depending on your service plans, you might have a set amount of usage resources per month or you might be billed for the resources that you consume. Your resource usage is calculated based on the number of model evaluations and explanations. Evaluations and explanations are measured in resource units.

See [IBM watsonx.governance plan options](#).

Learn more [↗](#)

- [What's new](#)
- [Use cases](#)
- [Feature differences between watsonx deployments](#)
- [Platform architecture](#)
- [Security of foundation models](#)
- [Services and integrations](#)
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